

(첨부2) 2026 금융 AI Challenge 기능명세서

팀명	주주의관제탑
구성원 성명	<제출자 직접 기재> — 개인(1인) 참가
서비스 명칭	주주의관제탑
배포 URL	https://jujutower.com (www.jujutower.com → 301 → 배포 URL)
작성일	2026-09-02

On this document. The 양식's five section headings are preserved verbatim in Korean; the body is written in English at the entrant's own choice. The product's user-facing surface is Korean only, so every quoted product string, field name and 공시 term below stays in Korean — a string inside 「」 is what the screen actually renders, transcribed from the running service, not a paraphrase.

Three notation rules, carried from 첨부1 and used throughout. A verified fact carries a [근거] source — a command, a 공시 접수번호, or the live URL that answers it; a derived or estimated figure is marked ▷ and is never presented as a measurement; and **every figure read off the live service is a dated snapshot**, because the corpus is refreshed twice a day and the board's numbers move with it. Where a snapshot is quoted, its date and time (KST) are quoted with it.

What this document is, and what 첨부1 is. 첨부1 is the plan and the argument. This document is the **specification of what is deployed and running**, and nothing else: §1-§4 describe the service as it runs at the URL above, and §5 is a procedure a judge can execute against that URL without contacting the entrant. It restates none of 첨부1's argument, and it claims **no users** — the service has been built, deployed and measured, never operated, so no usage, adoption or outcome figure appears here.

The 양식's own exclusion rule is applied strictly. 미구현 또는 향후 구현 예정 기능은 제외 — §1 and §2 list only what a judge can exercise today, and §1 names the excluded items explicitly so that a judge does not go looking for them.

1. MVP 구현 범위

Scope statement. Everything below is implemented, deployed and serving at https://jujutower.com as of the 2026-09-02 18:00 KST snapshot [근거: the live service]. The service runs as a production build in containers on the entrant's own server, behind nginx and Cloudflare; there is no staging URL, no demo mode and no dummy data — every disclosure fact on every screen comes from a filing stored in the service's own corpus.

1.1 Deployed surfaces

#	Surface	Route	What is implemented
1	관제 현황판 (the landing)	/	The whole-market board of shareholder rights with a deadline: the 2026 소멸 가치 headline, the 소멸주의보 strip with a live countdown, the four tabs (전체 · 유증 · CB · 매수청구) with whole-board counts, and the ranked D-day list, read 15 rows at a time

#	Surface	Route	What is implemented
2	내 종목 조회	<code>/stocks</code> , <code>/stocks/{corp_code}</code>	종목명/종목코드 search with a suggestion list, one issuer's 진행 중인 권리, per-holding conversion (배정 신주 · 초과청약 한도), and the 2026년 농친 돈 block with its own 집계 범위 statement
3	이벤트 상세	<code>/events/{rcept_no}</code>	One filing's facts, each with a <code>[근거]</code> control that opens the verbatim quoted span from the original document , the DART 원문 link, the schedule, the 발행 조건, and the 정정공시 반영 / 정정 이력
4	AI 질문	<code>/ask</code> (+ a desktop widget, + 질문 스트립 on 이벤트 상세)	A streaming, citation-forced agent over the verified corpus, printing each tool call as a one-line fact row and ending every answer with 근거 N건 · 접수번호 · 생성 시각 (KST). Anonymous, no login, no quota
5	계정	<code>/auth/login</code> , <code>/auth/reset</code>	계정 만들기 · 로그인 · 로그아웃 · 비밀번호 재설정 (emailed link) · 계정 삭제
6	보유 종목 (내 포트폴리오)	<code>/portfolio</code>	The account's holdings (담기 · 수정 · 삭제), 다가오는 마감 / 지나간 마감 with per-holding conversion, and the 채권 돈 mark
7	샘플 보유 종목	<code>/portfolio?sample=1</code>	An anonymous, one-click sample built from today's real filings , four holdings picked one per state, with example 보유량
8	알림 설정	<code>/portfolio/notifications</code>	수신 주소 (the account email) and the 시점 칩 7일 전 / 3일 전 / 1일 전 / 당일 (default 7일 + 1일); an empty selection is a valid setting meaning "no mail"
9	마감 임박 이메일	the reader's inbox + <code>/portfolio/notifications</code>	The daily send path, on its own 08:30 KST schedule: one mail per reader per deadline per chip, idempotent, with no won amount in the mail
10	비밀번호 재설정 메일	the reader's inbox + <code>/auth/reset</code>	The reset link mail, over the same real SMTP transport
11	의견 보내기	every reader page's footer	A message to the operator with no contact field — no email, no account, nothing stored on this side
12	검색·공유 표면	<code>/robots.txt</code> , <code>/sitemap.xml</code> , link previews	Per-route titles and descriptions, canonicals, a sitemap, Open Graph / share card, a web manifest, and inline JSON-LD
13	운영 관제	<code>/ops</code>	Implemented and deployed, operator-only , and 「 심사 대상 아님 」 — it is linked from nowhere in the reader chrome and is listed here only for completeness
14	상태 점검	<code>/api/health</code>	A liveness endpoint that reports <code>{"status":"ok", ...}</code> with the current KST time; it is what the external uptime monitor watches

1.2 The rights types in scope

Three rights types, all three serving:

	권리	What expires	Where the countdown comes from
①	유상증자 신주인수권	the 증서 lapses if it is neither exercised nor sold	the 신주인수권증서 매매기간, read out of the filing's prose by the reading layer and gated
②	전환사채 오버행	the 전환청구 window opens and the dilution lands	DART's structured fields only — zero model calls
③	주식매수청구권	the 반대의사 통지 deadline passes	DART's structured fields only — zero model calls

1.3 Deployment and data, as they run

- **Runtime.** Production build, six containers (web · API · worker · scheduler · Postgres · Redis) on the entrant's own server, reached through Cloudflare and a shared nginx edge. Only the web container is reachable from the edge; the database is on a project-internal network with no host port.
- **Corpus.** One store, **19 tables**, holding **1,359 events** with every filing version and a raw snapshot of every document, so any quoted span can be located in the document it came from [근거: the production database, seeded 2026-09-02] .
- **Refresh.** The collection pipeline runs **twice daily at 07:30 and 19:30 KST**; the notification send is a separate stage on its own schedule at **08:30 KST**.
- **Live board, 2026-09-02 18:00 KST snapshot** [근거: <https://jujutower.com/api/board>] : 감시 중 445건 (유증 12 · CB 422 · 매수청구 11) · 30일 이내 마감 40건 · 소멸 앞둔 신주인수권 15건 · headline 2026년 소멸한 신주인수권 가치 > 718.1억원 (밴드 하한 > 548.7억원). These four numbers move with every refresh; a judge will see that day's values, not these.
- **Search and share surface, same snapshot:** /robots.txt 200 (1,972 bytes, ending in Sitemap: <https://jujutower.com/sitemap.xml>), /sitemap.xml 200 with 800 URLs (3 static + 445 event pages + 352 issuer pages), all on the apex host.
- **Monitoring.** An external synthetic probe checks /api/health and the landing every 10 minutes and mails the operator on failure; a 17-check read-only smoke suite runs the whole chain in one command (§5.6).

1.4 미구현 · 향후 구현 예정 — excluded from this document by the 양식's rule

None of the following is described in §2-§5, and a judge should not look for it:

- **카카오톡 알림** — the row is visible on 알림 설정 with a bordered 「예정」 chip and **no working control**, deliberately: there is no server field behind it. Email is the only channel that sends.
- **마이데이터 연결 · 계좌 연동 · 거래/주문/체결** — none exists, and none is planned inside the MVP. Holdings are typed in by the reader; the service informs, and the reader acts in their own MTS.
- **권리 데이터 MCP 서버** — a roadmap item in 첨부1 §6, not built.
- **EB, 분할합병 · 주식교환·이전, 배당 기준일** — outside the MVP's three rights types.
- **가격 피드 / 시장 데이터 벤더** — not used anywhere; every amount is derived from the filings themselves.
- **운영 관제 (/ops)** — built, but operator-only and 심사 대상 아님.

2. 주요 기능 목록

Every row is **구현 완료 · 배포됨** — a feature that is only partly built belongs in §1.4's exclusion list, not here, so this table has no 「부분 구현」 state. **관련 화면** gives the route and the Korean surface name a judge will see on it.

기능명	기능 설명	관련 화면	구현 상태
관제 현황판	The market-wide board of every watched right with a deadline, ranked by D-day, with whole-board tab counts (전체 · 유증 · CB · 매수청구) and a 15-row display window that discloses the rest in place. Re-reads itself every 60 s and says so when the 기준시각 has moved.	/ — 관제 현황판	구현 완료 · 배포됨
2026년 소멸 가치 헤드라인	The retrospective total — ▷ 소멸한 신주인수권 가치, its conservative band lower edge, and the 배정 증서 / 소멸 비율 sentence. Every figure carries the 「추정」 tag on screen.	/ — 관제 현황판	구현 완료 · 배포됨
소멸주의보	The named hazard strip: how many 신주인수권 are counting down right now, the nearest 청약 마감 date, and a live ticking countdown to it. Names a count of issuers when several close on the same day.	/ — 소멸주의보	구현 완료 · 배포됨
내 종목 조회 (검색)	종목명 or 종목코드 resolution with a suggestion list of up to eight candidates while the reader types, each carrying its 종목코드. A bare submit resolves unique-or-declines; the service never silently opens a different company.	/stocks — 내 종목 조회	구현 완료 · 배포됨
진행 중인 권리	One issuer's live rights: the countdown, the window, the 배정비율, the 초과청약 비율, and the 발행가 확정 전 state with its 확정 예정일 — no invented amount.	/stocks/{corp_code} — 내 종목 조회	구현 완료 · 배포됨
보유량 환산	The reader types (or picks 100 · 500 · 1,000) a 보유 주식 수 and the page converts it into 배정 신주 and 초과청약 한도, printing the arithmetic — 「= 500주 × 0.0863800841 · 1주 미만 버림」. Anonymous input is session-memory only — 「서버 전송 없음」.	/stocks/{corp_code} — 내 종목 조회	구현 완료 · 배포됨
2026년 놓친 돈	For a lapsed ①: 발행 – 청약 = 소멸 증서 with the market-wide ▷ 이론가치, then the reader's own ▷ amount at their 보유량, with the 하한 beside it and the derivation printed. Marked 추정 on screen, and bounded by a stated 집계 범위.	/stocks/{corp_code} — 2026년 놓친 돈	구현 완료 · 배포됨
이벤트 상세 + 원문 인용	One filing's facts as field rows; each [근거] opens the verbatim span from the original document , and the DART 원문 link opens the filing itself. Footer: 「모든 값은 DART 공시에서만 나왔습니다 · 각 항목의 [근거]가 원문 구절과 접수번호로 연결됩니다」.	/events/{rcept_no} — 이벤트 상세	구현 완료 · 배포됨
정정공시 반영 · 정정 이력	The latest correction stated in one sentence (what changed, and whether the schedule moved), plus the version history behind it. A stale link still opens the right event after a 정정 changed the 접수번호.	/events/{rcept_no} — 정정공시 반영	구현 완료 · 배포됨

기능명	기능 설명	관련 화면	구현 상태
AI 질문 (에이전트)	An autonomous function-calling agent over seven server-side tools: it decides what to call, in what order, over as many rounds as it needs. Every tool call prints as a one-line fact row while it happens; the answer streams incrementally and ends with 「근거 N건 · {접수번호} · {생성 시각} KST」. Anonymous, unlimited, no login.	<code>/ask</code> — AI 질문; the desktop widget; 이벤트 상세's 질문 스트립	구현 완료 · 배포됨
검증된 계산 블록	A calculation the reader can audit: 「검증된 계산」 with each input labelled — the reader's own number tagged 「입력」, the disclosure-read value carrying its citation chip — then the expression and the 결과. The arithmetic is deterministic and the model does not perform it.	<code>/ask</code> — AI 질문	구현 완료 · 배포됨
계정	계정 만들기 · 로그인 · 로그아웃 · 비밀번호 재설정 (emailed link) · 계정 삭제. Stored PII is exactly email + password hash; deletion removes the email immediately and cascades.	<code>/auth/login</code> , <code>/auth/reset</code> — 로그인 · 비밀번호 재설정	구현 완료 · 배포됨
보유 종목 (담기 · 수정 · 삭제)	The account's holdings, each with its 진행 중인 권리 summary, split into 다가오는 마감 and 지나간 마감 with per-holding conversion. 「담기」 from an event page preselects the issuer and writes nothing until the reader saves.	<code>/portfolio</code> — 보유 종목	구현 완료 · 배포됨
챙긴 돈	The reader may mark a lapsed ① as 「청약·매도로 챙겼습니다」. It is stored as the reader's own assertion (filing number + timestamp, no amount) and is never mixed into disclosure data or any aggregate.	<code>/portfolio</code> — 지나간 마감	구현 완료 · 배포됨
샘플 보유 종목	One click, no signup: four holdings picked from today's real filings , one per state (① 발행가 확정 전 · ② 전환청구 개시 · ① 소멸 · ③ 통지 마감 지남), with example 보유량. Banner: 「샘플 보유 종목 — 구성 예시입니다. 종목·공시·마감은 실제, 계정·보유량은 예시입니다。」 Edits persist in the browser only.	<code>/portfolio?sample=1</code> — 샘플 보유 종목	구현 완료 · 배포됨
알림 설정	수신 주소 (the account email — there is no second address) and the 시점 칩 7일 전 / 3일 전 / 1일 전 / 당일, default 7일 + 1일. Deselecting every chip is a valid setting meaning no mail. Hidden entirely in 샘플 모드, because no address exists there.	<code>/portfolio/notifications</code> — 알림 설정	구현 완료 · 배포됨
마감 임박 이메일	The daily send, on its own 08:30 KST schedule against each reader's chips. Subject 「[주주의관제탑] {종목} — {마감명} {D-표기} ({date})」; the body carries 마감 · 기간 · 보유 (and 발행가 state for ①), the 보기 링크, the 접수번호 and an off-switch line. No won amount, ever . One mail per reader per deadline per chip, recorded only after the transport accepted it.	the reader's inbox + <code>/portfolio/notifications</code>	구현 완료 · 배포됨 (\$5.5)
비밀번호 재설정 메일	Subject 「[주주의관제탑] 비밀번호 재설정」, a single-use link with a stated expiry, over the same real SMTP transport.	the reader's inbox + <code>/auth/reset</code>	구현 완료 · 배포됨

기능명	기능 설명	관련 화면	구현 상태
의견 보내기	A message to the operator from any reader page. No contact field : no email, no account and no IP is sent, and the surface says so. Nothing is stored on this side.	every reader page's footer — 의견 보내기	구현 완료 · 배포됨
검색·공유 표면	<code>robots.txt</code> and a <code>sitemap.xml</code> generated from the live corpus, a per-route <code><title>/description</code> , per-route canonicals, Open Graph + share card, a web manifest and inline JSON-LD. Gated and operator surfaces are <code>noindex</code> , <code>nofollow</code> and disallowed.	<code>/robots.txt</code> , <code>/sitemap.xml</code> , any page's <code><head></code> , a link preview	구현 완료 · 배포됨
운영 관제	Six read-only operator tabs (개요 · 게이트 · 정확도 · 대화 로그 · 사용자 · 의견). Pure observation — no action in it can change what the product exposes. Linked from nowhere in the reader chrome.	<code>/ops</code> — 주주의관제탑 운영 (심사 대상 아님)	구현 완료 · 배포됨
상태 점검 · 가동 감시	<code>/api/health</code> answers <code>{"status":"ok", ...}</code> with the KST time; an external probe checks it and the landing every 10 minutes and mails the operator on failure.	<code>/api/health</code>	구현 완료 · 배포됨

3. 사용자 이용 흐름

The order in which a stranger meets the main features after opening the 배포 URL. **The product is mobile-first**: the 390-wide phone is a first-class path, not a reduced one — every step below was executed at 390 and at 1280 and renders without horizontal overflow (\$5.7).

3.1 The primary path — no account, no install

1. **랜딩 = 관제 현황판**. The reader opens `https://jujutower.com` and lands on a market-wide board, not a chat window and not a login wall. The first things on screen are the 2026 소멸 가치 headline and the 소멸주의보 strip with its ticking countdown to the nearest 청약 마감.
2. **A 종목 검색**. The hero's own field (「종목명 또는 종목코드 — 예: 계양전기」) suggests candidates as the reader types; choosing one opens **내 종목 조회** for that issuer.
3. **환산**. On that page the reader types their 보유 주식 수 (or taps 100 · 500 · 1,000주). The right's 배정 신주 and 초과청약 한도 appear with the arithmetic printed, and 「서버 전송 없음」 says where that number lives. If the issuer had a lapsed ① this year, **2026년 놓친 돈** states what was lost at that holding size.
4. **이벤트 상세**. 「상세 보기 →」 opens the filing's own page: every field with a [근거] that shows the **verbatim sentence from the original document**, the DART 원문 link beside it, and the 정정공시 반영 line if the schedule has moved.
5. **AI 질문**. From the event page's 질문 스트립, the nav slot, or the desktop widget. The reader presses one of four question cards (two of which name a company the server picks from today's corpus) or types their own. Tool rows appear one at a time as the agent reads; the answer streams and ends with 근거 N건 · 접수번호 · 생성 시각.
6. **샘플 보유 종목**. Pressing 「보유 종목」 in the nav while signed out — or 「샘플 포트폴리오로 둘러보기」 on the 로그인 page — opens, in one click, a portfolio of four **real, live** filings with example 보유량, so the 2층 can be understood

before any account exists.

7. **계정 만들기 → 담기**. Only now is an account offered, and only because it buys something the anonymous path cannot have: a saved holding and a deadline mail. 계정 만들기 takes an email and a password (8자 이상) and there is no verification mail in the way.
8. **알림 설정**. The 시점 칩 (7일 전 · 3일 전 · 1일 전 · 당일) decide when the mail arrives; the default is 7일 + 1일 and deselecting all of them means no mail.
9. **마감 임박 이메일**. At 08:30 KST the reader receives one mail per matching deadline — the facts, the 보기 링크 back into the product, the 접수번호, and the off switch. Never a won amount.

3.2 Alternate entries

- **A shared link**. `https://jujutower.com/events/{rcept_no}` is a complete surface on its own, so a filing forwarded in a chat opens on the facts with their citations. This is the most common way a stranger will arrive, and it needs no account.
- **The 소멸주의보 strip**. A reader who came for the headline can go straight from the strip into the ranked list, and from a row into that filing.
- **AI 질문 first**. `/ask` is in the nav from every page; on a phone it *is* the surface (there is no widget below 768).
- **A wrong address**. Any unmatched path or a filing the service does not expose renders the product's own Korean not-found — 「이 주소에 해당하는 공시가 없습니다」 — with the address echoed back and a way to the board. It never explains *why*, because the reason is internal.

4. AI 및 데이터 처리 방식

4.1 The AI's role — a three-layer split, stated as a specification

AI reads and speaks; determinism calculates. The boundary is fixed in code, not asked for in a prompt.

Layer 1 — reading. DART's structured fields give the skeleton only. What a shareholder actually needs is written in prose across the 증권신고서 / 주요사항보고서 body, so a schema-driven LLM extraction reads exactly **ten fields and no more**: ① 신주인수권증서 상장·매매기간 · 청약 취급처 · 실권주 처리 방식 · 초과청약 조건 · 발행가액 산정방법; ② 리픽싱 세부 조건 · 콜·풋 세부 스케줄 · 보호예수/전매제한 해제일; ③ 반대이사 통지 방법·절차; and, across all three, **정정 해석** (what changed and how the reader's D-day moved). Every extracted value must come back **with the verbatim sentence it came from**. Documents are never read whole: the body is cut by `<TITLE>` section and, for very large filings, windowed around the field's anchor.

Layer 2 — verification (deterministic, no model). Every field carries its own gate, and the gates judge against evidence the model never saw — the deterministically parsed label table and the stored API rows: arithmetic consistency, date ordering, comparison with the structured field, and — on **every** field — a **citation-span gate** that locates the model's quoted sentence character by character in the stored document. A verdict is one of `passed` / `failed(code)` / `tbd` / `not_evaluable(code)`, and a **check that could not run is never a pass**. A field that fails is not shown to the reader at all — it is recorded with a reason code. Measured on the stored corpus: **409 renderable field instances, 0 of them outside passed / tbd, 0 tbd fields leaking a value, 0 exposable events in a non-exposable state** [근거: the stored corpus's own gate run report — a single offline command, 0 OpenDART requests and 0 model calls, byte-identical across runs; 2026-08-20 corpus. The

project's separate exposure summary counts 418 field instances on a wider definition; the narrower 409 is the figure used here and in 첨부1].

Layer 3 — speaking. Sentences are generated only on top of verified data, and the AI 질문 agent is an autonomous function-calling agent over **seven** server-side tools (event search, event read, portfolio read, feedback, operator contact, an auditable calculator, and a prompt-injection guard) — it decides which to call, in what order, over as many rounds as it needs, and when to answer. Nothing is fetched before it speaks and no tool fires because a question matched a pattern. At the generation boundary, mechanically: a disclosure-fact sentence reaches the reader **only** if it rests on a verified verbatim span, every number in it came from a tool's own payload, and every quoted span is byte-identical to the stored document. Anything else is **discarded before it is streamed** — not flagged, not shown with a warning. **Refusal is a first-class answer** with five fixed categories (철회 · 확정 전 · 공시에 없음 · 검증 미통과 · 계산 요청), and a refusal is citation-forced too: asked for an amount that is 확정 전, the agent gives every known fact with its 근거 and refuses **the amount only**.

What the AI does not do. All money and D-day arithmetic is **LLM-free** — 배정 신주, 단수주, 소멸 가치, the KST countdown, the 보호예수 해제일 derivation — and is unit-tested. The 철회 judgement is deterministic (a row shape in the 정정사항 table, not a keyword). **No model training of any kind is involved**; the 양식 asks how a 생성형 AI 모델 is *applied*, and that is the whole of this section.

Operating envelope. One reading model, `gemini-3.7-flash`, used in exactly two places: offline schema extraction and the in-request agent turn. The agent runs at a low reasoning setting and logs a ▷ **cost ledger** per turn — ▷ **\$0.0093 per turn** on a measured sample, an estimate and never a billed figure. Each pipeline run carries hard request/call ceilings and stops on exceeding them. The serving path **never calls OpenDART** (it reads stored rows only) and reaches the model **only** through the agent; both rules are enforced by code scans, not by convention.

4.2 Data used

Every source is public and free. There is no price feed and no market-data vendor.

데이터	출처	쓰임새
공시 정형 상세	DART OpenAPI (유상증자결정 · 전환사채권 발행결정 · 회사합병 결정 등)	The skeleton fields: 배정기준일, 발행가, 배정비율, 전환가액·전환청구기간, 매수청구 접수기간
공시 본문 원문	DART documents (ZIP → XML): 주요사항보고서 · 증권신고서 · 증권발행실적보고서 · 정정공시	The source for prose extraction and the ground truth every citation span is located in
청약 결과	증권발행실적보고서	발행 증서 / 증서 청약 / 초과청약 / 실권주 — the document the 소멸 총액 is framed on
정정 이력	the filing's 3. 정정사항 table	What changed, when, and how the reader's D-day moved
사용자 데이터	typed in by the reader : 보유 종목·수량, 계정 이메일, 알림 시점 칩	Personalisation and the deadline mail. No 마이데이터, no brokerage link, no order flow

Stored shape. One store, **19 tables**. The corpus chain is `corp → event → filing_version → snapshot`, with every observed 접수번호 kept as a version and every document snapshotted with its raw body, so a quote can always be re-located in the exact bytes it was read from. Extraction rows carry the model, the prompt/schema version, the

token counts, the ▷ cost and the gate verdicts. Excluded filings are **retained with a suppression reason**, never deleted, so an exclusion is auditable.

4.3 Input and output data, per surface

Surface	Input data	Output data
관제 현황판	none (anonymous)	Counts, D-day rows, the ▷ 소멸 가치 headline, the 기준시각
내 종목 조회	a 종목명 / 종목코드; optionally a 보유 주식 수 (anonymous: browser session only, never sent)	진행 중인 권리, 배정 신주 · 초과청약 한도 with the arithmetic printed, ▷ 놓친 돈 at that holding size
이벤트 상세	a 접수번호 (in the URL)	The filing's verified fields, each with its verbatim quoted span and the DART 원문 link; the correction story
AI 질문	a free-text question; the anonymous tab handle	A streamed answer, the tool fact rows, the citation blocks, an auditable calculation block, 근거 N건 · 접수번호 · 생성 시각 (KST)
보유 종목	종목 + 보유 주수 (account-bound); a 채긴 돈 mark	Per-holding rights, D-day sections, ▷ 놓친 돈 / 채긴 돈
알림 설정	the 시점 칩 selection	The stored preference; nothing else
마감 임박 이메일	(no reader input) the account's holdings × chips × today's deadlines	One mail: 종목 · 마감명 · D-표기 · 기간 · 보유 주수 (and the 발행가 state for ①) · the 보기 링크 · the 접수번호 · the off switch. No won amount

4.4 개인정보 · 민감정보 processing

What is stored, and nothing else. A reader account is exactly **email + password hash** (+ timestamps) — no name, no phone, no brokerage link, no market identity, no activity trail. Beside it: the reader's **holdings** (종목 고유번호 + 주수), the **notification chips**, and a **채긴 돈 mark** (filing number + timestamp, **and no amount**). Passwords are stored as a hash only; the plaintext never persists.

- **민감정보 (금융거래정보 포함) is not collected.** No account number, no balance, no transaction history, no 마이데이터 connection, no brokerage link. Holdings are a number the reader typed.
- **Anonymous state never reaches the server.** The anonymous 환산 input lives in the browser session (「서버 전송 없음」 is printed beside it) and the sample portfolio's edits live in the browser's own storage. Migration into an account is **offered, never automatic**; declining sends nothing.
- **AI 질문 conversations are stored anonymously, and it is a schema property, not a policy.** The conversation tables carry **no account, email, IP or user-agent column and no foreign key in either direction**, and the session handle is randomly minted rather than derived from anything about the reader. The surface tells the reader this in its own words rather than claiming the conversation disappears.
- **Deletion is immediate and total.** 계정 삭제 removes the email at once and cascades to sessions, reset grants, holdings, claims and preferences.
- **The mails carry the minimum.** Only two kinds of mail exist — the deadline alert the reader configured and the password reset — and there is **no marketing or digest mail**. Every mail carries an off-switch line, and **no mail carries a won amount** (the 발행가 line states a *state*, never a figure).

- **Everything is first-party.** The service runs on the entrant's own server; reader data is not shared with, or sold to, any third party. Measured on the live pages: **no page contacts a third-party origin** — no analytics, no external font or script, no beacon; the only external references in the HTML are the DART 원문 links themselves.
- **The operator door is separate.** /ops has its own credential with no relation to any reader table, and the operator surface itself shows a **count** of a reader's holdings, never their contents.

5. MVP 검증 방법

How to read this section. Each block is a step a judge performs unaided on the 배포 URL, written as **URL → 조작 → 예상 결과**, where 예상 결과 quotes a Korean string the surface actually renders. Blocks marked **[익명]** need no account at all. Blocks marked **[계정]** need one; §5.1 says how to get it.

One structural warning, stated up front. This service runs on **live disclosure**, so *which* companies appear changes from day to day. Every block below is therefore written as an **observable state** ("an ① whose 발행가 is not yet fixed, counting down"), and every concrete company or number in it is a **dated example from the 2026-09-02 18:00 KST snapshot**, given only so the expected shape is unambiguous. A judge will see different names and the same states. That is the product working, not drifting.

5.1 테스트 계정

- **Primary — self-signup, instant, no credential handed out.** Open <https://jujutower.com/auth/login>, press 「계정 만들기」, and enter any email address the judge controls and a password of **8자 이상**. There is **no verification mail in the way**: the account exists immediately and 로그인 routes straight to 보유 종목. This is the recommended path — it lets each judge hold their own account, and it means no shared password is printed in a public document. For the account-bound mail blocks (§5.5) the address must be one the judge can actually read.
- **Secondary — an account prepared by the operator.** If a pre-made 심사용 계정 is preferred, the entrant supplies its credentials separately rather than in this document. **This is an open decision at the time of writing**, and §5.5's mail blocks are the only ones for which it would matter.
- **Deleting the account afterwards** is one control: 알림 설정 → 「계정 삭제」, which erases the email immediately.

5.2 샘플 입력값

#	Input	Value to use	Notes
S1	A live 종목 with an upcoming deadline	Take the top row of the board on the day of judging. Dated example (2026-09-02): 틀젠 — 신주인수권증서 매매 마감 2026-09-07 , then D-5	Any row of the ranked list works; an ① row is the richest
S2	A 보유 주식 수	500 (or the 500주 chip)	Dated example result for 틀젠: 배정 신주 43주 , 초과청약 한도 +8주
S3	A 접수번호 for 이벤트 상세	The 상세 보기 link from S1. Dated example: 20260806000329	Also reachable directly at <code>/events/{rcept_no}</code>

#	Input	Value to use	Notes
S4	The 샘플 포트폴리오	https://jujutower.com/portfolio?sample=1	Four holdings, one per state , picked from today's filings; example 보유량 500 / 300 / 500 / 100주
S5	An AI 질문	Press one of the four question cards on /ask — they are the product's own sentences. Dated example: 「아이에이 유상증자, 1,000주 보유 시 배정 신주는 몇 주인가요?」	Two of the four cards name a company the server picks from today's corpus, so the sentence itself changes daily
S6	A bad 접수번호	https://jujutower.com/events/00000000000000	Must be a Korean 404 page, not a server error

5.3 검증 절차 — the anonymous blocks

V1 · 관제 현황판 [익명]

- URL: <https://jujutower.com/>
- 조작: open it; read the headline, then the 소멸주의보 strip, then the ranked list; tap a tab.
- 예상 결과: the headline 「2026년에 소멸한 신주인수권 가치」 with a figure carrying the 「추정」 tag and 「밴드 하한 … (권리락 조정 가정)」 beneath it; a summary line 「감시 중 N건 · 30일 이내 마감 N건」; the 소멸주의보 sentence 「지금도 N건의 신주인수권이 소멸을 앞두고 있습니다 (가장 빠른 청약 마감 YYYY-MM-DD, N개 종목).」 above a **counter that visibly ticks down every second**; four tabs (전체 · 유상증자 신주인수권 · 전환사채 오버행 · 주식매수청구권) whose counts stay whole-board when a tab is selected; and a ranked list of rows, each with 종목명, 마감명, the date and a D-표기. A 「N건 더 보기」 control at the foot of the list discloses the next rows in place. *Dated example (2026-09-02 18:00 KST):* ▷ 718.1억원 · 감시 중 445건 · 30일 이내 마감 40건 · 소멸주의보 15건, 가장 빠른 청약 마감 2026-09-04, 3개 종목.

V2 · 내 종목 조회와 보유량 환산 [익명]

- URL: <https://jujutower.com/> (the hero field), or <https://jujutower.com/stocks>
- 조작: type the S1 종목명 into 「종목명 또는 종목코드 — 예: 계양전기」; pick the suggestion; on the issuer's page press the 500주 chip.
- 예상 결과: the suggestion list shows the company **with its 종목코드** while typing; choosing it opens /stocks/{고유번호} titled with the 종목명. The page shows 「진행 중인 권리 — N건」 and, for the right, its 마감명, D-표기, the date with 「KST」, and the window. After 500주: 「보유 500주」, 「배정비율 (1주당)」, 「배정 신주」 with the printed arithmetic 「= 500주 × {배정비율} · 1주 미만 버림」, and 「초과청약 한도」. Beside the input: 「서버 전송 없음」. If the 발행가 is not yet fixed the page says 「발행가 확정 전 · 확정 예정 {날짜} — 확정 후 증서 이론가치와 금액을 환산합니다」 and shows **no money at all** — that is the product refusing to invent an amount, not a missing feature. *Dated example: 툐젠 · 배정비율 0.0863800841 · 배정 신주 43주 · 초과청약 한도 +8주 · 발행가 확정 전 (확정 예정 2026-09-11).*

V3 · 2026년 놓친 돈 [익명]

- URL: the same page, for an issuer that had a **lapsed** ① this year. Reachable in one click from the 샘플 포트폴리오's 「놓친 돈 상세 →」 (V6), which always contains one. Dated example: <https://jujutower.com/stocks/01409022> (페니트리움바이오).

- **조작:** press 500주.
- **예상 결과:** a 「2026년 놓친 돈」 block headed 「청약도 매도도 하지 않았다면, 2026년 이 종목에서 사라진 가치」, with the market-wide derivation printed as 「발행 {N}주 – 청약 {N}주 = 소멸 {N}주 ({N}%) · {금액}억원추정」, then the reader's own amount at 500주 with a **하한** beside it and the derivation 「배정 {N}주 = 500주 × 배정비율 {…} (1주 미만 버림) · 증서 1주 이론가치 = 확정발행가 × 할인율 ÷ (1 – 할인율)」. Every figure carries the 「추정」 tag; a 집계 범위 note and the sentence 「모든 값은 DART 공시에서만 나왔습니다 · 보유량 환산은 공시된 배정비율과의 곱셈이며, 시장 가격을 사용하지 않습니다」 close the page. *Dated example:* 소멸 1,336,556주 (15.76%) · ▷ 13.7억원, and at 500주 ▷ 79,182원 (하한 ▷ 68,546원), 배정 77주 × ▷ 1,028원.

V4 · 이벤트 상세와 원문 인용 — the load-bearing verification [익명]

- **URL:** <https://jujutower.com/events/{접수번호}> (S3), or 「상세 보기 →」 from V2.
- **조작:** read the countdown block; press any **[근거]** ; then open 「DART 원문 ↗」 in a new tab and search the filing for the quoted sentence.
- **예상 결과:** the page is titled 「{종목명} — {마감명}」 and carries 「접수번호 {번호}」, 「최초 공시 {날짜}」 and — when the filing has been amended — 「정정 반영」. Pressing **[근거]** opens the **verbatim sentence from the original document**, which a judge can find character-for-character in the DART filing itself. This is the load-bearing check of the whole service: every value on the page is either a DART structured field or a model reading that survived its gate, and the quote is the proof. The 정정 block states in one sentence what changed and whether the schedule moved. Footer: 「모든 값은 DART 공시에서만 나왔습니다 · 각 항목의 [근거]가 원문 구절과 접수번호로 연결됩니다」.

V5 · AI 질문 [익명]

- **URL:** <https://jujutower.com/ask>
- **조작:** press **one** of the four question cards (S5), and watch the screen while it answers.
- **예상 결과:** 「안녕하세요!」 and 「주주의 권리를 지키기 위해 공시를 근거로 질문에 답합니다」 above four cards. After pressing one, **the answer arrives incrementally, not as one late block**: the tool rows appear **one at a time** as the agent works — 「이벤트 검색 {종목}」 → N건 · …, 「이벤트 읽기 → …」, 「계산 → …」 — then a 「공시에서 읽은 값」 block, then the prose, each disclosure-fact sentence carrying a numbered citation marker, and finally the footer 「근거 N건 · {접수번호} · {YYYY-MM-DD HH:MM} KST」 with 「DART 원문 {접수번호} ↗」 links. If the card asked for a calculation, a 「검증된 계산」 block shows each input labelled — the reader's own number tagged 「입력」, the disclosure-read value carrying its citation — then the expression and the 결과. No login is asked for, and no quota is shown anywhere. *Executed 2026-09-02 18:15 KST:* 「아이에이 유상증자, 1,000주 보유 시 배정 신주는 몇 주인가요?」 → three tool rows, a 검증된 계산 block ($1,000\text{주} \times 0.507594018 = 507\text{주}$), and 「근거 2건 · 20260818000250 · 2026-09-02 18:15 KST」, in 5 visibly distinct screen states over ~10 s.

V6 · 샘플 보유 종목 [익명]

- **URL:** <https://jujutower.com/portfolio?sample=1> (equally: press 「보유 종목」 in the nav while signed out, or 「샘플 포트폴리오로 둘러보기」 on the 로그인 page — both open the same surface)
- **조작:** open it; read the four rows, then 다가오는 마감 and 지나간 마감; press 「수정」 on a row, change the 보유량, save, and reload the page.
- **예상 결과:** the banner 「샘플 보유 종목 — 구성 예시입니다. 종목·공시·마감은 실제, 계정·보유량은 예시입니다」; **four holdings, four different companies**, with a 「기준 {날짜} (KST)」 line; and the four **states** — in 다가오는 마감 an ① with a live D-표기 and 「발행가 확정 전」 plus an ② 전환청구 개시 row with its 오버행 · 전환 시 주식수 · 전환가액; in

지나간 마감 an ① whose 증서 기간 has passed, carrying 놓친 돈 with the 「청약·매도로 챙겼습니다」 control, and a ③ 통지 마감 지남 row. An edited 보유량 survives the reload (it is kept in the browser, not on the server). **The companies are today's, so expect different names** — assert the four states, not the names. *Dated example (2026-09-02): 아이에이 ① D-43 (발행가 확정 전, 배정 신주 253주) · 제이에스링크 ② D-1 · 페니트리움바이오 ① 기간 지남 D+37 (놓친 돈 ▷ 79,182원) · 휴맥스 ③ 통지 마감 지남 D+6.*

V7 · The honest 404 [익명]

- URL: `https://jujutower.com/events/00000000000000`
- 조작: open it.
- 예상 결과: HTTP 404 (never a 500), rendering the product's own Korean page: 「이 주소에 해당하는 공시가 없습니다。」 / 「관제 현황판에서 감시 중인 공시를 확인하실 수 있습니다.」, the address the judge typed echoed back, and 「관제 현황판으로 →」. It deliberately does **not** say why — "this filing exists but we are not showing it" would leak exactly what the exposure contract exists to prevent.

V8 · Origin, redirects and the search surface [익명]

- URLs: `http://jujutower.com/`, `https://www.jujutower.com/`, `https://jujutower.com/robots.txt`, `https://jujutower.com/sitemap.xml`, `https://jujutower.com/api/health`
- 예상 결과: `http://` → 301 → `https://`; `www.` → 301 → the apex, **path and query preserved**; `/robots.txt` 200 ending in Sitemap: `https://jujutower.com/sitemap.xml`; `/sitemap.xml` 200 listing every event and issuer page on the apex host (800 URLs in the 2026-09-02 snapshot) and **no** `/ops`, `/auth` or `/portfolio` URL; `/api/health` 200 with `{"status":"ok", ...}` and the current KST time. Any page's `<head>` carries its own title, description, self-canonical and `og:image`.

5.4 계정 기반 블록 [계정]

These need the account from §5.1. They are stated with the same expected strings; the entrant's own verification of them is recorded in §5.8.

A1 · 계정 만들기 · 로그인. `https://jujutower.com/auth/login` → 「계정 만들기」→ email + 비밀번호 (8자 이상) → the account exists immediately and the reader lands on 보유 종목. A duplicate address answers 「이미 가입된 이메일입니다 — 로그인해 주세요.」; a short password answers 「비밀번호는 8자 이상이어야 합니다.」; wrong credentials answer 「이메일 또는 비밀번호가 일치하지 않습니다.」 — one message for both fields, on purpose.

A2 · 담기. From an event page press 「보유 종목에 담기 →」 (or 종목 추가 on 보유 종목), enter a 보유 주수, save. The holding appears with 「계정에 저장 · 마감 알림의 기준」 and its 진행 중인 권리 summary; 다가오는 마감 shows its countdown converted at that holding size. Following the link alone writes nothing.

A3 · 알림 설정. `https://jujutower.com/portfolio/notifications` — opened **without** a session it redirects to 로그인 (it is gated; `/portfolio` itself does **not** redirect — signed out it renders the 샘플 보유 종목 instead); opened with one, 「마감 임박 이메일」 shows the 수신 주소 (the account email) and the chips 「7일 전 · 3일 전 · 1일 전 · 당일」, with 7일 전 and 1일 전 preselected. Deselecting every chip is a **valid** setting meaning no mail. The KakaoTalk row is present with a bordered 「예정」 chip and **no working control** — 「준비되면 이 자리에서 켤 수 있습니다」.

A4 · 비밀번호 재설정 메일. `/auth/login` → 「비밀번호 재설정」→ enter the address → the surface answers 「재설정 링크를 보냈습니다 — 메일함을 확인해 주세요.」 and a real mail arrives, subject 「[주주의관계탐] 비밀번호 재설정」, carrying a single-use link with a stated expiry.

A5 · 마감 임박 이메일. With a holding whose deadline is 7 / 3 / 1 / 0 days away and the matching chip set, the 08:30 KST run sends one mail per matching deadline. Subject 「[주주의관제탑] {종목} — {마감명} {D-표기} ({date})」; the body carries 마감 · 기간 · 보유 (and, for an ①, the 발행가 **state**: 「확정 전 (확정 예정일 …)」), then 「주주의관제탑에서 보기 →」 with the link, then 출처: 공시 접수번호 and the off-switch line. **No won amount appears anywhere in the mail** — the same rule as the screen's 확정발행가 전 금액 금지, kept structurally.

5.5 The mail channel, stated exactly

Status: 구현 완료 · 배포됨, with the transport proven live on production — a real password-reset mail was sent from the production stack over the configured SMTP account on **2026-09-02** [근거: the production API log]. What has **not** happened yet is a deadline mail reaching a reader *from production*: the production database currently holds no account whose holdings have a deadline 7 / 3 / 1 / 0 days out, so the daily selection correctly finds nothing to send. The send path itself was proven end to end against a local SMTP sink (① · ② deadline mails, idempotency, the off switch, the ceiling). This document therefore describes the mail as **deployed and transport-proven, not as demonstrated to a reader from production**, which is the honest distinction.

5.6 One-command reproducible check

The entrant maintains a single command that exercises the whole chain — Cloudflare → nginx → the web container → the API — from a laptop with nothing installed (no dependency step, standard library only). It is included here because it states, in executable form, exactly what the service asserts about itself; the entrant can run it on request, and V1-V8 above cover the same ground by hand:

```
make smoke-prod          # or: python3 scripts/smoke_production.py
```

17 read-only checks, ~8 seconds, no dependencies, no account, no writes, and **no model call**: `/api/health` (asserting the body, not just the status) · the landing with its security headers · the `www` 301 · the `http` 301 · the board with real rows · one 이벤트 page and one 종목 page built from the board's own rows · a bad 접수번호 answering **404 and not 500** · the AI 질문 start cards · the `/ops` door · `robots.txt` · `sitemap.xml` · the web manifest and its icons · the share image · `noindex` on the gated surfaces · **no off-origin script or link on the landing** · and the three neighbouring sites on the same box still answering 200. A non-zero exit means at least one failed, and the failing check is named.

5.7 브라우저 · 런타임 제약

- **Any modern browser.** Verified in **Google Chrome 152** on macOS; Safari and other current Chromium browsers are equally supported. No extension, no plug-in, **no install of any kind**.
- **HTTPS only.** `http://` and `www.` both redirect; the apex is canonical.
- **Two viewports, both first-class.** Verified at **1280** (desktop) and **390** (phone) — every surface in §5.3 renders at 390 with **no horizontal scrolling**. Below 768 the AI 질문 page replaces the desktop widget; that is the design, not a degradation.
- **JavaScript is required**, as for any modern web application. AI 질문 streams over Server-Sent Events, so a corporate proxy that buffers responses may make the answer appear all at once instead of progressively; the answer itself is unaffected.
- **All dates and countdowns are KST**, and are labelled 「KST」 on screen.
- **No account is needed** for V1-V8. Nothing in the anonymous path writes data.

5.8 이 MVP의 한계 — stated plainly

1. **The corpus refreshes twice a day (07:30 / 19:30 KST).** A filing published in the last few hours may not be on the board yet. When the latest measurement is older than its threshold the board **says so itself**, in a banner — 「데이터가 갱신되지 않고 있습니다. 아래 값은 기준 시각의 공시 기준이며, 그 이후의 정정공시는 반영 전일 수 있습니다.」 — above a 「기준 {날짜} {시각} KST · N시간 전 데이터」 line. That banner is the product being honest about its own freshness, and a judge may well see it.
2. **An ① before its 발행가 is fixed shows no money.** The 확정발행가 is typically published about one business day before 청약, so the most urgent events show 배정 주수 and a 「발행가 확정 전」 chip instead of an amount. This is a deliberate rule (확정발행가 전 금액 금지), applied on screen **and in the mail**.
3. **The companies on every live surface change daily**, including the 샘플 포트폴리오's four rows and two of the four AI 질문 cards. That is why §5 is written by state rather than by 종목.
4. **Money figures are estimates and are marked as such.** 놓친 돈 and the 소멸 가치 headline carry the 「추정」 tag on screen and ▶ in this document; the 증서 이론가치 is derived by inverting each filing's **own** 발행가 산식 — there is no price feed — and it is not a realised profit or loss. The screen says so: 「실제 손익은 개별 청약·매도 행동에 따라 다릅니다」.
5. **놓친 돈 is bounded by a stated 집계 범위** (유증 2026-01-01부터 · CB 2025-06-01부터). Anything before that window is not counted, and the page says so rather than implying zero.
6. **Three ② fields (리픽싱 · 콜·풋 · 보호예수) are read by the model** and are subject to the same gate as everything else; a field that fails its gate is **not shown at all**, so an occasional absent row is the trust rule working, not a rendering bug.
7. **The deadline mail arrives only at 08:30 KST, and only for a chip that matches.** It is not instant, and it is not sent for a deadline further out than the reader's chips.
8. **The reported extraction accuracy is a cross-model judgement.** 첨부1 quotes **98.6 % strict** (213/216, 95 % Wilson [96-100 %]) — and that figure must never be quoted without the clause that belongs to it: **the labels were produced by one model judging another model's extractions (Claude judging Gemini), not by human ground truth**. It is stated here for the same reason it is stated there.
9. **Withdrawn filings answer by 접수번호 but are not found by company search** — the intended consequence of the exposure contract.
10. **Cloudflare is in front of the origin.** A 52x page is Cloudflare reporting that it could not reach the origin; it is an infrastructure state, not a product screen.
11. **There are no users.** The service is built, deployed and measured, never operated. No usage or outcome figure is claimed here or anywhere else.

5.9 검증 실행 기록 (entrant's own run, 2026-09-02)

V1-V7 were each executed once against `https://jujutower.com` by the entrant on **2026-09-02, 18:05-18:25 KST**, in **real Google Chrome 152** (headful, a throwaway profile, driven over the DevTools protocol), at **1280** and at **390**; every Korean string quoted above is one that was observed on screen, and the dated examples are that run's own readings (V5's question was asked **once**, at 1280; the answered thread was then read again at 390). V8's status codes and redirects were checked by direct HTTP request against the same origin on the same day, as was the one-command check in §5.6. The **[계정]** blocks in §5.4 and the mail in §5.5 are verified by the operator, who holds the production account and the mailbox; the anonymous script is complete without them.

The live figures in this document are a **2026-09-02 18:00 KST snapshot** of a service whose numbers move twice a day; estimates are marked ▷ on the page as 「추정」 and here as ▷; facts carry a [근거] — a command, a 공시 접수번호, or the URL that answers them. The judge-facing procedure is §5; the plan and the argument behind the service are 첨부1 공모전 기획서.